Snowmobiling & Winter Recreation Safety Guide

Safety essentials for snowmobiling, snowshoeing, and winter backcountry travel in Pierce County

Pierce County's Cascade Mountains transform into a winter playground with hundreds of miles of snowmobile trails and backcountry terrain. But winter conditions are the most unforgiving environment we operate in. Our SAR unit runs snowmobiles, tracked quads, and a SnoCat for winter missions — we know firsthand how fast conditions can turn deadly.

WINTER IN THE CASCADES

Snow depths in the Pierce County Cascades regularly exceed 10-15 feet. Whiteout conditions can develop in minutes. Temperatures at elevation routinely drop below zero with wind chill. Avalanche terrain is everywhere above treeline. Winter backcountry travel demands serious preparation and respect.

Snowmobile Areas Near Pierce County

Greenwater / Cascade Crest

The primary snowmobile staging area for Pierce County riders. Access via SR 410 from Enumclaw. Extensive trail system through national forest land with varying difficulty. Sno-Park permit required. Check WSDOT for pass conditions and chain requirements.

Chinook Pass Area

SR 410 closes at the summit in winter, opening up miles of unplowed road and surrounding terrain for snowmobiling. Higher elevation means deeper snow and more alpine exposure. Avalanche terrain is prevalent — know before you go.

Crystal Mountain / White Pass Backcountry

Backcountry access near ski areas. Popular with experienced riders seeking powder and alpine terrain. Significant avalanche exposure. Only for riders with avalanche training and proper rescue gear.

Avalanche Safety

Avalanches kill more recreationists in Washington than any other natural hazard. If you travel above treeline or in steep terrain (30-45 degree slopes), avalanche education is not optional — it's survival.

- ☐ Check the forecast: Northwest Avalanche Center (nwac.us) daily. Know the danger rating and problem types.
- ☐ Carry the gear: Avalanche beacon (transceiver), probe, and shovel. ALL THREE. Practice using them.
- ☐ Take a course: AIARE Level 1 at minimum. Many local organizations offer courses each winter.
- ☐ Travel one at a time: Cross avalanche-prone slopes individually. Watch each person from a safe spot.
- ☐ Know the terrain: Slopes 30-45 degrees are the most dangerous. Convex rolls, wind-loaded faces, cornices.
- ☐ Recognize red flags: Recent avalanche activity, whumpfing sounds, shooting cracks, rapid warming, heavy new snow.

SAR PRO TIP

Burial time is everything. The survival rate drops from 90% at 15 minutes to under 30% at 35 minutes. Your partners are your first responders — organized rescue takes too long. Every member of your group must carry a beacon, probe, and shovel, and KNOW HOW TO USE THEM.

Snowmobile Safety

- ☐ Registration: WA Sno-Park permit and ORV registration required.
- ☐ Helmet: DOT-approved required. Full-face recommended for warmth and protection.
- ☐ Clothing: Insulated snowmobile suit or layered system. Wind protection is critical at speed.

- ☐ Emergency kit: Fire starter, space blanket, extra food/water, headlamp, basic tools, spare belt if applicable.
- ☐ Buddy system: NEVER ride alone in the backcountry. Groups of 3+ recommended (one injured, one stays, one goes for help).
- ☐ Speed: Reduce speed in flat light, unfamiliar terrain, and near trees. Tree wells are deadly traps.
- ☐ Fuel: Know your range. Cold burns more fuel. Carry a siphon and know your group's fuel situation.

Snowshoeing & Winter Hiking

Snowshoeing has boomed in popularity but many participants underestimate winter conditions. The trail you followed in may be unrecognizable on the return if snow is falling. GPS navigation is essential — trail markers can be buried under feet of snow.

- ☐ Carry traction (snowshoes or crampons) appropriate to conditions
- ☐ Gaiters to keep snow out of boots. Waterproof insulated boots.
- ☐ Extra insulation layers — you'll be warm while moving but cold when stopped
- ☐ Thermos with hot liquid. Calories are critical in cold.
- ☐ Navigation: GPS with offline maps is essential. Trail markers disappear under snow.
- ☐ Start early. Winter daylight is short (8-9 hours). Budget extra time for breaking trail.

SAR PRO TIP

Tree wells — the void around the base of a tree where snow hasn't compacted — are one of winter's hidden killers. Fall into one headfirst and you can suffocate in minutes even in shallow snow. Stay well clear of tree bases, especially on steep terrain. If someone falls in, dig them out immediately — it's a time-critical rescue.

Winter Emergency Kit

In addition to the standard Ten Essentials, winter travel demands extra preparation. Hypothermia can set in during any stop if you're wet from exertion.

- ☐ Extra insulating layers (puffy jacket that fits OVER your shell)
- ☐ Chemical hand/toe warmers (multiple pairs)
- ☐ Fire starting kit in waterproof container
- ☐ Emergency bivy or space blanket
- ☐ High-calorie food that won't freeze solid (nut butter, chocolate, cheese)
- ☐ Insulated water bottle or thermos (regular bottles freeze quickly)
- ☐ Headlamp with lithium batteries (perform better in cold)
- ☐ Satellite communicator — cell service is rarely available in winter backcountry

SATELLITE 911 TEXTING

T-Mobile offers satellite 911 texting — FREE for all carriers (emergency 911 only). Full satellite messaging is $10/mo. Winter backcountry has essentially ZERO cell coverage. Sign up at t-mobile.com/coverage/satellite-phone-service/911-texting-signup